

Lecture 5 Processing of Ceramic Parts - Pressing

A wide variety of fabrication processes are used to prepare ceramics powder for ceramic product. The process for a particular product is based on the material, shape complexity of the product, property requirements and cost. Processing of ceramics generally involves three basic steps:

- Ceramic Powder Preparation (crushing/milling/grinding)
- Mixing powder particles with additives (to impart special characteristics)
- Shaping, drying and firing the material

Ceramic fabrication processes can be divided into six common categories as shown in figure 1.

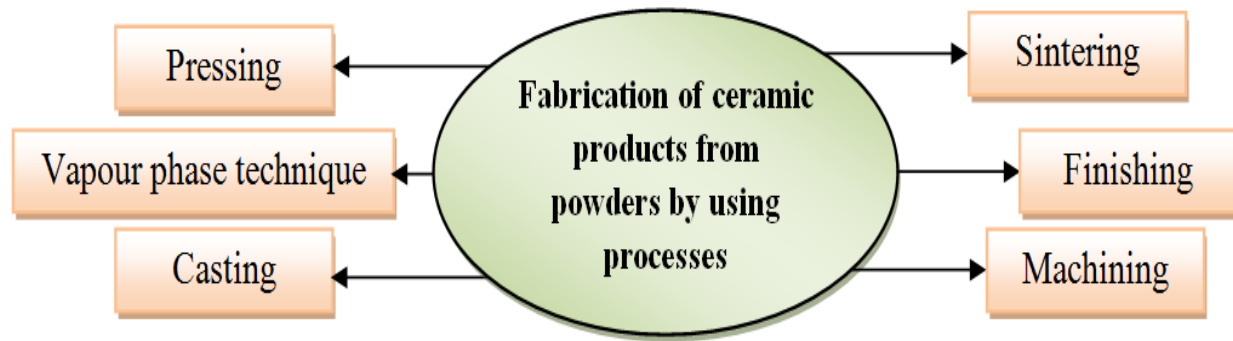


Figure 1 Various processes used in fabricated ceramic products

Pressing

Pressing is one of the simplest methods of ceramic powder processing for development of ceramic based products. Raw material is crushed in fine powders which is mixed with additives and then processed into useful products. In pressing operation, powder containing little amount of water is compacted under pressure. A wide range of traditional and advanced ceramic based products are processed by powder pressing method which includes electronic and electrical ceramic items, floor and wall tiles and insulator of spark plug. Powder pressing can be carried out in number of ways which are as follows (figure 2):

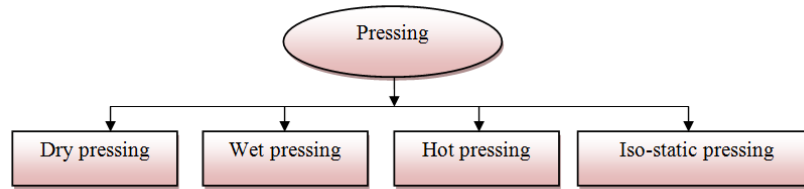


Figure 2 Powder pressing techniques

Dry pressing

Dry pressing technique is used for simple shapes such as abrasive products and whitewares. The schematic of dry pressing technique is shown in figure 3. Water content in the powder mixture is very low (less than 4%). Various binders (organic as well as inorganic) may be added in the mixture depending on the requirement. Production rate is high in dry pressing method and close dimensional tolerances are achieved. The advantages associated with dry pressing technique are maximum production rate and better tolerance control. The disadvantages of this process include non-uniformity in density and wear resistance of dies.

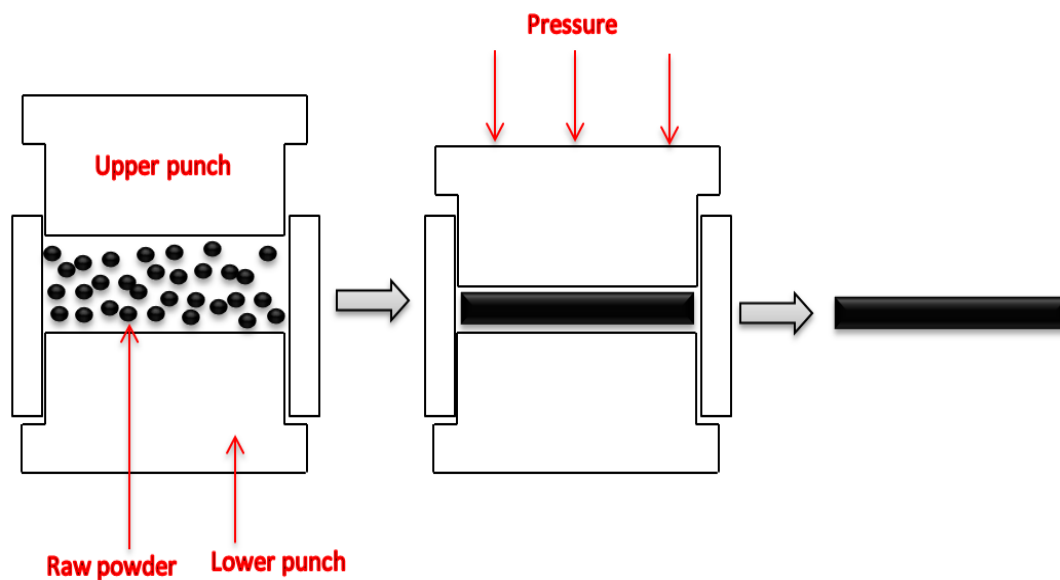


Figure 3 Dry pressing technique

Wet and hot pressing

In wet pressing method, product is processed under high pressure in a mold. Moisture content is relatively high (10-15%). In wet pressing technique, production rate is high and it is possible to handle intricate shapes but the process is suitable for smaller jobs and there is not better control over dimensional accuracy.

In hot pressing method, both pressure and temperature are applied which reduce the void content of the part and produces a denser and stronger product. The advantage of hot pressing technique is that strong and dense parts can be comfortably processed. The demerits of this process are the requirement of controlled atmosphere and shorter life of dies.

Iso-static pressing

Iso-static pressing is used to obtain uniform density in the product. Insulators of spark plug are fabricated by iso-static pressing method. Powder mixture is placed around a central mandrel pin in a flexible mold on which fluid pressure is applied from outside. The schematic of iso-static pressing is shown in figure 4. There is uniform density distribution in parts process with iso-static pressing but this process requires high infrastructural cost.

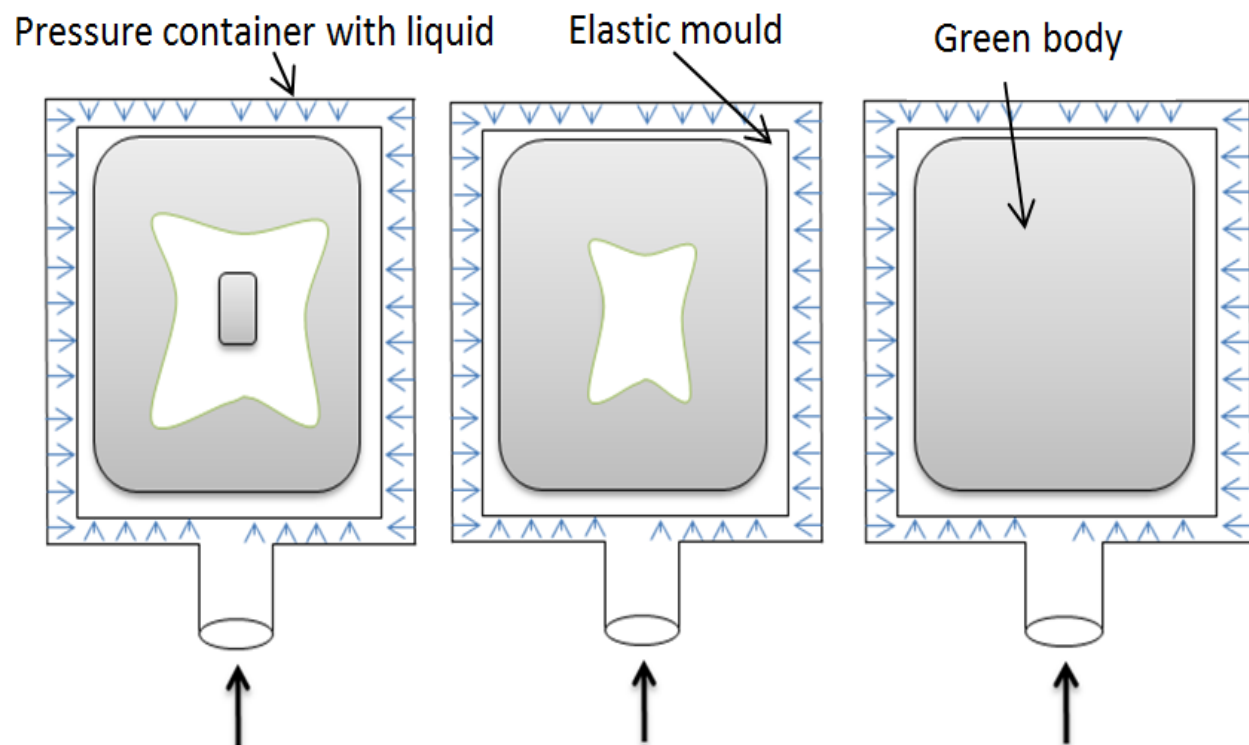


Figure 4 Iso-static pressing technique

Lec-6: Processing of Ceramic Parts – II

This chapter continues the discussion on processes of ceramic product fabrication. The chapter covers the discussion on casting, sintering and vapour phase techniques.

Casting

Casting is the ceramic processing technique in which powders are cast into useful shapes. The process selection is based on the powder material being processed, the shape desired and the property requirements of the part. Treatment of the starting powders is usually required for slip and tape casting, and the treatment depends on the product. Fabrication of products from ceramics powder based on casting processes includes extrusion, jiggering, die pressing, and injection molding.

Slip casting

A SLIP is a suspension of ceramic particles in a liquid. A porous mould made of plaster of paris (POP) is used in slip casting. The slip must have optimum fluidity to flow easily into the mold. The slip is poured into the mould and the capillary action of the porous mould draws water from the slip and forms a solid layer on the inside of the mould. When the desired part thickness has been obtained, the excess slip is drained out. The part is partially dried in the mould to cause it to shrink away from the mould and develop adequate rigidity for handling. The slip is continuously supplied to replenish the absorbed water for solid ceramic parts. The sequence of operations in slip casting is schematically shown in figure 1.

This process is an economical way to form complex shapes and is used for hollow objects such as art ware and sanitary ware. Slip preparation is critical to slip casting and requires optimization among a number of factors, including viscosity, solids contents, slip stability, casting rate, drying rate, and drying shrinkage.

Pressure casting has begun to replace conventional slip casting. In pressure casting, pressure is applied to the slip to supplement or replace the capillary action of the porous mold. The advantages of pressure casting over conventional slip casting are higher casting rate, ability to automate the process, and extension of the useful application of slip casting to colloidal nonclay powders.

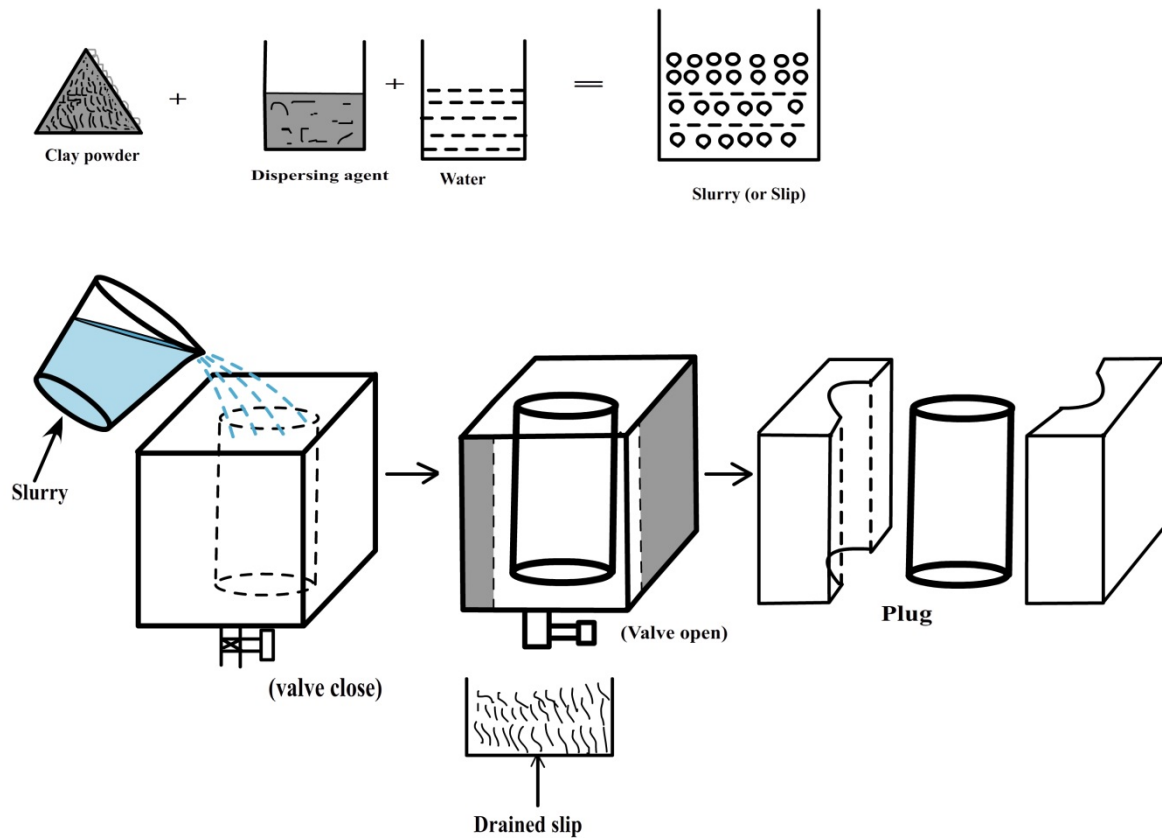


Figure1: Sequence of operations in slip-casting process using ceramic powder

Sintering

In sintering process, ceramic powder mixture is heated at sufficiently high temperature in a controlled atmosphere. Bonding of the particles of powder mixture takes place at high temperature. The type and strength of the bond depends on several factors such as recrystallization temperature, grain growth, diffusion mechanism etc. The governing factors in sintering process are sintering time, temperature and atmosphere of heating chamber. Mechanism of sintering process is complex which depends on various variables. There are two popular mechanisms which occur during sintering operation as (Figure 2):

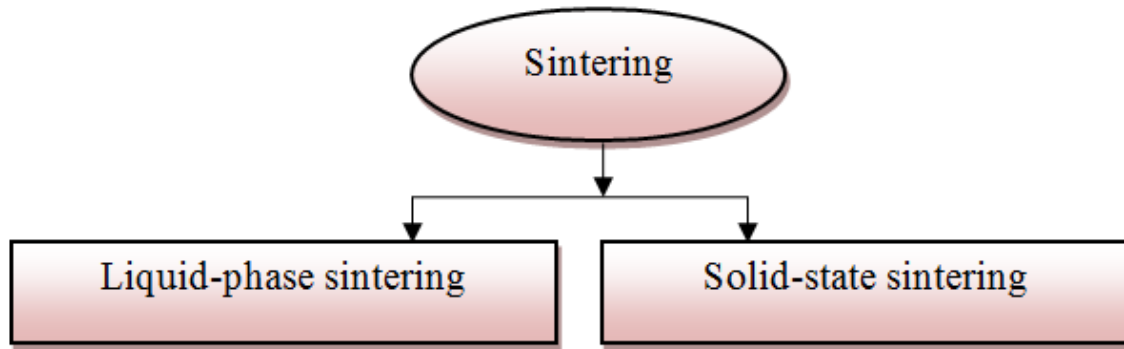


Figure 2 Sintering mechanisms

In liquid phase sintering process, bonding can take place at interface of two particles in liquid phase. When the powder mixture is heated, it forms a viscous liquid which promotes densification by viscous flow. A lot of traditional ceramic products such as dinnerware, sanitary ware, and electrical porcelain are processed through liquid state sintering.

In solid-state sintering process, bonding between the particles takes place through diffusion process. Solid-state sintering mechanism is predominant mechanism in many advanced ceramic products.

Vapour phase techniques

Vapour phase techniques are becoming popular and finding its place in many applications. There are two basic vapour phase processes which are (Figure 3):

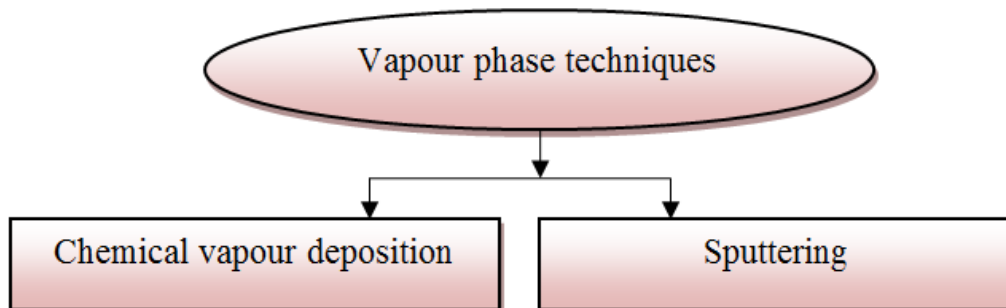


Figure 3 Vapour phase processes.

In chemical vapour phase technique, ceramics are brought in contact with a heated surface through gases containing atoms where gases react and form a coating. The process is widely used to make a ceramic coating on metals and carbide tools. Wear resistance coatings are also developed with this method.

Sputtering is another method to achieve ceramic coatings. Reactive sputtering is the most commonly used sputtering method. In this technique, materials to be coated are vaporized from target material by the application of magnetron. These vaporized atoms are deposited on a substrate (material to be coated) at a constant temperature where a reaction takes place on the substrate to form coating.